

Maternal meat and fat consumption during pregnancy and suspected atopic eczema in Japanese infants aged 3-4 months: the Osaka Maternal and Child He...



Interest has increased in the possibility that maternal dietary intake during pregnancy might influence the development of allergic disorders in children. The present prospective study examined the association of maternal intake of selected foods high in fatty acids and specific types of fatty acids during pregnancy with the risk of suspected atopic eczema among Japanese infants aged 3-4 months. Subjects were 771 mother-child pairs. Information on maternal dietary intake during pregnancy was assessed with a validated self-administered diet history questionnaire. The term 'suspected atopic eczema' was used to define an outcome based on results of our questionnaire completed by mothers 3-4 months postpartum. The risk of suspected atopic eczema was 8.4% (n = 65). Higher maternal intake of meat during pregnancy was significantly associated with an increased risk of suspected atopic eczema in the offspring: the multivariate odds ratio (OR) for the highest vs. lowest quartile was 2.59 [95% confidence interval (CI): 1.15-6.17, p for trend = 0.01]. The positive association was strengthened when the definition of the outcome was confined to a definite physician's diagnosis of atopic eczema (n = 35): the multivariate OR between extreme quartiles was 3.53 (95% CI: 1.19-12.23, p for trend = 0.02). No material exposure-response relationships were observed between maternal intake of eggs, dairy products, fish, total fat, saturated fatty acids, monounsaturated fatty acids, n-3 polyunsaturated fatty acids, alpha-linolenic acid, eicosapentaenoic acid, docosahexaenoic acid, n-6 polyunsaturated fatty acids, linoleic acid, arachidonic acid and cholesterol and the ratio of n-3 to n-6 polyunsaturated fatty acid consumption and the risk of suspected atopic eczema. Higher maternal meat intake may increase the risk of infantile atopic eczema, whereas we found no evidence that maternal intake of fish and n-3 polyunsaturated fatty acids are preventive against infantile atopic eczema. (c) 2009 John Wiley & Sons A/S